

TNBC Application - Descriptive Summary

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- Loading necessary libraries

```
library(SpatialExperiment)
library(spdep)
library(adespatial)
library(ggplot2)
library(ggpubr)
library(dplyr)
```

```
#number of tissue
length(unique(colData(spe)$sample_id))
```

```
[1] 39
```

Number of cells per tissue

```
table(colData(spe)$sample_id)
```

Sample_01	Sample_02	Sample_03	Sample_04	Sample_05	Sample_06	Sample_07	Sample_08
5199	3033	5671	5381	4252	4894	3308	3786
Sample_09	Sample_10	Sample_11	Sample_12	Sample_13	Sample_14	Sample_15	Sample_16
5105	4066	4615	5976	6755	5251	3470	6815
Sample_17	Sample_18	Sample_19	Sample_20	Sample_21	Sample_23	Sample_24	Sample_25
6726	5000	3371	4067	5148	3416	3911	2290
Sample_26	Sample_27	Sample_28	Sample_29	Sample_30	Sample_31	Sample_32	Sample_33
5409	3621	5917	4332	9886	2781	4380	1716
Sample_34	Sample_35	Sample_36	Sample_37	Sample_39	Sample_40	Sample_41	
3091	7754	2482	5244	3663	3933	3479	

```
max(table(colData(spe)$sample_id))
```

```
[1] 9886
```

```
min(max(table(colData(spe)$sample_id)))
```

```
[1] 9886
```

```
#find cells in Sample_04, Sample_05, Sample_26, Sample_39  
table(colData(spe)$sample_id)[c("Sample_04", "Sample_05", "Sample_26", "Sample_39")]
```

```
Sample_04 Sample_05 Sample_26 Sample_39  
      5381      4252      5409      3663
```

Expression analysis

```
sum(assay(spe) < 0)
```

```
[1] 0
```

Let's compute mean, median, min, max for each marker in each tissue, separately. Then, plot the results.

```
library(pheatmap)  
  
# Extract data  
expr <- assay(spe, "exprs")  
samples <- colData(spe)$sample_id  
  
# Compute mean expression per marker per tissue  
sample_levels <- unique(samples)  
  
mean_mat <- sapply(sample_levels, function(s) {  
  rowMeans(expr[, samples == s, drop = FALSE], na.rm = TRUE)  
})
```

```

# Set row/column names
rownames(mean_mat) <- rownames(expr)
colnames(mean_mat) <- sample_levels

# Color: 0 = white, low = yellow, high = red
my_colors <- colorRampPalette(c("white", "yellow", "red"))(100)

# Plot heatmap (no scaling)
pheatmap(mean_mat,
          scale = "none",
          color = my_colors,
          cluster_rows = FALSE,
          cluster_cols = FALSE,
          fontsize = 12,          # overall size
          fontsize_row = 12,      # marker labels (rows)
          fontsize_col = 12,
          filename = "Figure_D18.png",
          width = 7,
          height = 8)

```

Thresholding

```

# Threshold for the markers
thresholds <- c(
  betaCatenin = 0.253063192,
  CD11b = 1.100536222,
  CD11c = 0.213556566,
  CD138 = 1.038502303,
  CD16 = 0.304460727,
  CD20 = 0.296460556,
  CD209 = 0.406669859,
  CD3 = 0.250762424,
  CD31 = 1.006381212,
  CD4 = 0.223762,
  CD45 = 0.327450495,
  CD45RO = 0.236334141,
  CD56 = 0.770183826,
  CD63 = 0.268126101,
  CD68 = 1.128510101,
  CD8 = 0.260237333,

```

```

dsDNA = 0.20301697,
EGFR = 1.069264889,
FoxP3 = 0.34649899,
H3K27me3 = 0.201893232,
H3K9ac = 0.406071229,
HLA_Class_1 = 0.23773137,
HLADR = 0.4408116,
IDO = 0.312824202,
Keratin17 = 0.291843697,
Keratin6 = 0.282480556,
Ki67 = 0.210413737,
Lag3 = 1.239097091,
MPO = 0.324934545,
p53 = 0.265704293,
panKeratin = 0.244129571,
PD1 = 0.972859909,
PDL1 = 1.352024101,
pS6 = 0.282699091,
SMA = 0.250358024,
Vimentin = 0.28239596
)

threshold_vec <- thresholds[rownames(spe)]

# Check for any missing thresholds
if (any(is.na(threshold_vec))) {
  stop("Missing thresholds for: ",
       paste(rownames(spe)[is.na(threshold_vec)], collapse = ", "))
}

# Extract assay matrix
expr_mat <- assay(spe, "exprs")

# Apply thresholds row-wise and create binary matrix
binary_mat <- expr_mat >= threshold_vec
storage.mode(binary_mat) <- "integer" # convert TRUE/FALSE to 1/0

# Store as a new assay in spe
assay(spe, "binary_exprs") <- binary_mat

```

Binary marker summary

```
bin_mat <- assay(spe, "binary_exprs")
samples <- colData(spe)$sample_id

# Count number of present cells per marker per sample
sample_levels <- unique(samples)

count_mat <- sapply(sample_levels, function(s) {
  rowSums(bin_mat[, samples == s, drop = FALSE], na.rm = TRUE)
})

# Set row/column names
rownames(count_mat) <- rownames(bin_mat)
colnames(count_mat) <- sample_levels

# Plot heatmap
pheatmap(count_mat,
  scale = "none",
  cluster_rows = FALSE,
  cluster_cols = FALSE,
  color = colorRampPalette(c("white", "yellow", "red"))(100),
  fontsize = 12,          # overall size
  fontsize_row = 12,      # marker labels (rows)
  fontsize_col = 12,
  filename = "Figure_D19.png",
  width = 7,
  height = 8)
```

Binary markers - percentage cells

Percentage of cells each marker present in each tissue

```
library(pheatmap)

bin_mat <- assay(spe, "binary_exprs")
samples <- colData(spe)$sample_id

sample_levels <- unique(samples)
```

```

# Compute percentage of cells with marker present
perc_mat <- sapply(sample_levels, function(s) {
  idx <- samples == s
  rowMeans(bin_mat[, idx, drop = FALSE], na.rm = TRUE) * 100
})

# Set row/column names
rownames(perc_mat) <- rownames(bin_mat)
colnames(perc_mat) <- sample_levels

# Plot heatmap
pheatmap(perc_mat,
  scale = "none",
  cluster_rows = FALSE,
  cluster_cols = FALSE,
  color = colorRampPalette(c("white", "yellow", "red"))(100),
  fontsize = 12,
  fontsize_row = 12,
  fontsize_col = 12,
  filename = "Figure_D19.png",
  width = 7,
  height = 8)

```